

ANUSHA PAMALI DSILVA

CONTACT



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<https://github.com/Anushadsilva>

TECHNICAL SKILLS

- Programming & Scripting: Python, SQL, R, PowerShell
- Data Engineering & Processing: ETL/ELT Pipelines, Apache Spark, Apache Airflow, Pylceberg, PyArrow, Data Modeling, Data Lakes, Schema Evolution, Data Quality Validation
- Cloud Platforms: AWS (S3, Glue, Redshift, IAM, Lambda, Aurora), Azure (Azure SQL Database, API Management, DevOps), GCP (BigQuery)
- Databases & Warehousing: Azure SQL Database, Amazon Redshift, PostgreSQL, Cassandra, SQL Server (SSMS)
- APIs & Integration: REST APIs (Flask, FastAPI), OAuth 2.0, API Management
- BI & Visualization: Power BI, ERD Design
- DevOps & Tools: GitHub Actions, Docker, CI/CD Pipelines

CERTIFICATIONS

- AWS Certified Data Engineer – Associate, 01/2026
- Credential URL:
https://www.credly.com/badges/05132cae-3e58-4aec-a079-cb7d9c20ce54/public_url
- Data Engineering Nanodegree, Udacity, 08/2021
- Bootcamp – Data Engineering, Zac Wilson, 03/2025

EDUCATION

May 2023

Master's Information Systems
University of Utah

December 2018

Master's Software Engineering
Birla Institute of Technology and Science

May 2014

Bachelor's Computer Application
Mangalore University

Experienced Data Engineer with 8+ years at the intersection of software and data engineering, specializing in designing and delivering scalable, cloud-native data pipelines and automation solutions. Adept at enhancing data quality, security, and accessibility using Python, SQL, AWS, and Azure. Proven track record of driving operational efficiency and business value by building robust, insight-driven systems for enterprise clients. Strong engineering foundation combined with a strategic, data-first approach to solving complex data challenges.

EXPERIENCE

May 2023 - May 2025

Data Engineer *Rio Tinto*

- Architected and developed a Flask API-driven data pipeline to ingest transformer oil data into Azure SQL Database, secured via Azure API Management. Integrated CI/CD pipelines using GitHub Actions for automated deployments, collaborating with cross-functional teams to ensure seamless integration and alignment with business needs.
- Designed and optimized ETL workflows, increasing operational efficiency by 85% and significantly reducing costs.
- Implemented ETL solutions for the Operational Data Platform, enabling real-time data availability in the data warehouse and BI layer, adding 25+ KPIs to improve productivity reporting and decision-making by 98%.
- Conducted API testing with Postman and worked with cross-functional partners to validate JSON payloads and ensure data integrity. Configured secure SSL certificates, reducing data retrieval times by 50%.
- Extracted oil data from legacy source systems using ODBC connections, configured SSIS packages for scheduled data loads from source to target, improving efficiency by 90% and driving cost savings through generated reports.
- Created Power BI reports for KPIs and automated workflows with Power Automate, streamlining productivity tracking and enabling actionable insights.
- Resolved data issues in condition monitoring reports by modifying or writing complex SQL queries, views and stored procedures.

June 2022 - January 2023

Data Science and Engineering Intern *Rio Tinto*

- Created a dashboard to predict engine health based on oil sample data, reducing the cost and frequency of failures by 85%.
- Resolved issues in the existing ETL setup for extracting oil data and implemented an alternate solution that significantly improved performance and efficiency by 90%.
- Developed Python scripts to establish ODBC connections to source data, efficiently extracting and pushing delta records to the destination, ensuring continuous data synchronization and updates.
- Designed and optimized complex SQL queries to split downtime records by shift and accurately determine the production day.

May 2014 - August 2021

Senior Software Engineer *Wipro Limited*

- Built a utility in Python to handle reimbursements for customers wrongly charged on their credit card thereby increasing process efficiency to 88%.
- Developed Python scripts to encrypt files containing customers details and store it in S3.
- Automated policy creation for life insurance product using Python and improved processing time by 85%.
- Analyzed insurance product usage data in R Studio to get insights on products extensively used by customers.
- Built and maintained data extraction tool using SQL queries and Python libraries (pandas) to extract and consolidate data from different third-party vendors.
- Developed and executed SQL queries to retrieve information adhering to business requirements.
- Documented various insurance and credit card flows to ease training and development for new hires in the team.
- Collaborated with cross-functional teams using Agile methodologies, such as Scrum and Kanban, to deliver high-quality software products within tight deadlines.